

Slides_Lectu
re1_ERM...

Advanced Data Sciences

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Lecture 1, M2 - Maths/Info & PSND
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About the lecture

- 2 main parts : 32 hours (16h with Hebiri and 16h with Lacombe)
- Hebiri : Statistical analysis of ML algorithms
- Lacombe : Computational analysis for solutions approximation
- Evaluation :
 - Mainly (~15 over 20 points) based on a defense based on understanding of articles – Selection of articles soon available
 - ~5 over 20 points are obtained by solving some exercises

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What is Machine Learning?

The emphasis of machine learning is on devising automatic methods that perform a given task based on what was experienced in the past

Here are several examples :

- optical character recognition** : categorize images of handwritten characters by the letters represented
- topic spotting** : categorize news articles (say) as to whether they are about politics, sports, entertainment, etc.
- medical diagnosis** : diagnose a patient as a sufferer or non-sufferer of some disease
- customer segmentation** : dividing a customer base into groups of individuals that are similar in specific ways relevant to marketing (such as gender, interests, spending habits)
- fraud detection** : identify credit card transactions (for instance) which may be fraudulent in nature

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Typical tasks of machine learning 1

Supervised learning :

Data : (feature vector₁, label₁), ..., (feature vector_n, label_n)

- Prediction** : use historical data for predicting future values
- Classification** : identifying to which of a set of classes a new observation belongs, on the basis of a training set of data containing observations whose class membership is known
- Ranking** : construction of ranking models for information retrieval systems

Unsupervised learning :

Data : feature vector₁, ..., feature vector_n

- Outlier detection** : detecting a few observations that deviate so much from other observations as to arouse suspicion that it was generated by a different mechanism
- Clustering** : grouping a set of objects into homogeneous groups based on some similarity measure
- Dimensionality reduction** : mapping of data to a lower dimensional space such that uninformative variance in the data is discarded

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Typical tasks of machine learning 2

Semi-supervised learning :

Data : (feature vector₁, label₁), ..., (feature vector_n, label_n)
feature vector₁, ..., feature vector_n

- Prediction** : use historical data for predicting future values
- "Cheap features/costly labels"** : features are much easier to get than the associated labels, therefore $N \gg n$

On-line learning :

Data : (feature vector₁, label₁), ..., (feature vector_n, label_n)

- Prediction** : use historical data for predicting future values
- Sequential data** : it is preferable to use prediction methods that are easy to update after each round t

Active learning :

Data : (feature vector₁, label₁), ..., (feature vector_n, label_n)

- Prediction** : use historical data for predicting future values
- Focusing on "difficult features"** : at each round t , the user can choose feature_t, the nature reveals label_t

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Most popular methods of ML

	Supervised	Unsupervised
Easy to interpret	- logistic regression - linear regression - decision trees (CART)	hierarchical clustering
Highly accurate	- support vector machines - boosting (adaboost) - random forest - neural nets	- K-means (medoids) - EM for Gaussian mixtures

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Supervised learning : base notions

Notations principales

Features et labels

- Observations : n couples $Z_i = (X_i, Y_i)$ assumed *iid* and drawn from P which is unknown. We write $Z_i = (X_i, Y_i) \stackrel{\text{iid}}{\sim} P$
- The X_i s belong to the feature space $\mathcal{X}$. Typically, $\mathcal{X} = \mathbb{R}^d$ for some (large) integer d
- The Y_i s belong to the label space $\mathcal{Y}$. Typically, $\mathcal{Y}$ is finite or $\mathcal{Y}$ is a subset of $\mathbb{R}$

One main goal of supervised learning : predict the label Y associated to a new instance X , assuming that the couple (X, Y) is a new observation from the law P that is independent from the previously collected observations

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Main notations

Prediction algorithm and loss function

- A prediction function is a (measurable) function $g : \mathcal{X} \rightarrow \mathcal{Y}$
- $\mathcal{F}(\mathcal{X}, \mathcal{Y}) :=$ set of all prediction functions
- $\{Z_1, \dots, Z_n\}$ is the training set
- An algorithm is any function that takes a training set to produce a prediction function

$$G : \bigcup_{n=1}^{\infty} \mathcal{Z}^n \rightarrow \mathcal{F}(\mathcal{X}, \mathcal{Y}); \quad \mathcal{Z} = \mathcal{X} \times \mathcal{Y}$$

- Let $\ell(y, y')$ be the resulting loss when the true label is y and the predicted label is y' . The function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$ is the **cost function**.

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Main notations

Bayes predictor

- The quality of a prediction function $g : \mathcal{X} \rightarrow \mathcal{Y}$ is measured through its **risk** (or generalization error) :

$$R_P(g) = \mathbb{E}_P[\ell(Y, g(X))] = \int_{\mathcal{X} \times \mathcal{Y}} \ell(y, g(x)) dP(x, y) \quad (1)$$

- The «best» prediction function is the (or more precisely, a) function of $\mathcal{F}(\mathcal{X}, \mathcal{Y})$ minimizing the risk R_P :

$$g^* \in \arg \min_{g \in \mathcal{F}(\mathcal{X}, \mathcal{Y})} R_P(g) \quad (2)$$

- Such a function g^* might not exist but exists for usual loss functions

- This «best» function is called **Bayes predictor**. It depends on the unknown probability P and is then unknown itself

- Excess risk** : $\mathcal{E}_P(g) = R_P(g) - R_P(g^*)$.

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Exemple 1 : Binary classification

- $\mathcal{Y}$ contains only 2 elements, $\mathcal{Y} = \{0, 1\}$
- 0 – 1 loss : $\ell(y, y') = \mathbb{1}(y \neq y')$
- Framework : (binary) classification problem
- A Bayes classifier is given by

$$g^*(x) = \mathbb{1}(\eta^*(x) \geq 1/2) \quad (3)$$

where $\eta^*(x) = P(Y = 1|X = x) = \mathbb{E}[Y|X = x]$

- Exercise** : Show that $\mathcal{E}_P(g) = \mathbb{E}[|g(X) - g^*(X)|] \cdot [2\eta^*(X) - 1]$ *no Forestweck*

$$\text{Tot: } \mathbb{E}[|g|] = \mathbb{E}[\mathbb{E}[|g(X)|]]$$

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Exemple 2 : Least squares regression

- $\mathcal{Y} = \mathbb{R}$ or any interval in $\mathbb{R}$
- Quadratic loss : $\ell(y, y') = |y - y'|^2$
- More generally, any ℓ_p loss function might be used, with $p \geq 1$ and $\ell(y, y') = |y - y'|^p$

- A Bayes predictor, for $p = 2$, is given by

$$g^*(x) = \eta^*(x) = \mathbb{E}[Y|X = x] \quad (4)$$

- Exercise** : Show that $\mathcal{E}_P(g) = \mathbb{E}[|g(X) - g^*(X)|^2]$. *same !*

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Links between classification et regression – 1

- Whenever $\mathcal{Y} = \{0, 1\}$, the problem can be seen equivalently either as a classification or a regression problem (since $\mathcal{Y} \subset \mathbb{R}$)
- The corresponding Bayes predictors are

$$g^*(x) = \mathbb{1}(\eta^*(x) \geq 1/2); \quad \eta^*(x) = \mathbb{E}[Y|X = x]$$

- Assuming a prediction function η is built, by the **plug-in** principle, we are able to define the classifier

$$g(x) = \mathbb{1}(\eta(x) \geq 1/2)$$

- Question** : Assume the excess-risk (for the regression) of η is small. Does it imply that the excess-risk (for the classification) of g is also small?

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Links between classification et regression – 2

Recall

$$g^*(x) = \mathbb{1}(\eta^*(x) \geq 1/2)$$

For any regression function $\eta : \mathcal{X} \rightarrow \mathbb{R}$, define the classifier

$$g_\eta = \mathbb{1}(\eta \geq 1/2)$$

Theorem

Let $R_{cl,P}$ and $R_{reg,P}$ the classification and the (least squares) regression risks respectively; that is, $R_{cl,P}(g) = P[Y \neq g(X)]$ and $R_{reg,P}(\eta) = \mathbb{E}_P[(Y - \eta(X))^2]$. Then

$$R_{cl,P}(\hat{g}_n) - R_{cl,P}(g^*) \leq 2 \left(R_{reg,P}(\hat{\eta}_n) - R_{reg,P}(\eta^*) \right)^{1/2} \quad (5)$$

If the reg excess-risk is small $\Rightarrow$ the classif risk for classif is small.

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Algorithm : definitions

- Recall that an algorithm G is a mapping

$$G : \bigcup_{n=1}^{\infty} \mathcal{Z}^n \rightarrow \mathcal{F}(\mathcal{X}, \mathcal{Y})$$

- Then, $G = \{g_n : n \in \mathbb{N}_+\}$ where each $g_n : \mathcal{D}_n \mapsto \mathcal{F}(\mathcal{X}, \mathcal{Y})$ with $\mathcal{D}_n = \{Z_1, \dots, Z_n\}$

- Since $\mathcal{D}_n$ is random, $\hat{g}_n(x)$ is also random for all $x \in \mathcal{X}$

- Then $R_P(\hat{g}_n) = \int_{\mathcal{X} \times \mathcal{Y}} \ell(y, \hat{g}_n(x)) dP(x, y)$ is random as well

- Formal written :

$$R_P(\hat{g}_n) = \int_{\mathcal{X} \times \mathcal{Y}} \ell(y, \hat{g}_n(x)) dP(x, y)$$

- We can for instance study its expectation w.r.t. $\mathcal{D}_n \stackrel{\text{iid}}{\sim} P$, denoted by $\mathbb{E}_P[R_P(\hat{g}_n)]$

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Consistency of algorithms – 1

- An algorithm is said consistent w.r.t. the distribution P iff.

$$\mathbb{E}_P[R_P(\hat{g}_n)] \xrightarrow{n \rightarrow \infty} R_P(g^*)$$

- An algorithm is said consistent w.r.t. a family $\mathcal{P}$ of distributions iff. it is consistent w.r.t. any $P \in \mathcal{P}$

- An algorithm is said **universally consistent** iff. it is consistent w.r.t. any probability distribution P on $\mathcal{Z}$

- Universal consistency does not say much on the amount of data n needed to get a guarantee of the type

$$\mathbb{E}_P[R_P(\hat{g}_n)] \leq R_P(g^*) + \epsilon \quad \text{for a fixed } \epsilon > 0$$

- To get such a result we should show **uniform universal consistency**, that is

$$\lim_{n \rightarrow \infty} \sup_P \left\{ \mathbb{E}_P[R_P(\hat{g}_n)] - R_P(g^*) \right\} = 0$$

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Consistency of algorithms – 2

Theorem (No free-lunch Theorem)

If $\text{Card}(\mathcal{X}) = +\infty$, there exists no algorithm that is uniformly universally consistent, neither in least squares regression, nor in binary classification

- This negative result impose us to re-define the notion of «good» algorithm as a universally consistent algorithm that enjoys uniform convergence properties for a class of (relevant) distributions

- More precisely, if $\mathcal{P}$ is a set of distributions on $\mathcal{Z}$ that, we believe, contains P . We would like that the «good» algorithm satisfies

$$\lim_{n \rightarrow \infty} \sup_{P \in \mathcal{P}} \left\{ \mathbb{E}_P[R_P(\hat{g}_n)] - R_P(g^*) \right\} = 0$$

- and also that the sequence $\sup_{P \in \mathcal{P}} \left\{ \mathbb{E}_P[R_P(\hat{g}_n)] - R_P(g^*) \right\}$ is decreasing as fast as possible to 0

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Empirical risk minimization (ERM)

Introducing ERM

- Let $g : \mathcal{X} \rightarrow \mathcal{Y}$ and consider a loss function ℓ . The risk of g is given by :

$$R_P(g) = \mathbb{E}_P[\ell(Y, g(X))]$$

- We estimate $R_P(g)$ by the empirical risk of g

$$\hat{R}_n(g) = \frac{1}{n} \sum_{i=1}^n \ell(Y_i, g(X_i)) = \hat{\mathbb{E}}_n[\ell(Y, g(X))]$$

- Classical convergence guarantees if $\mathbb{E}_P[\ell(Y, g(X))^2] < +\infty$

- For instance LLN : $\hat{R}_n(g) \xrightarrow[n \rightarrow \infty]{P} R_P(g)$

- Introduce the ERM

$$\hat{g}_{n,G} \in \arg \min_{g \in \mathcal{G}} \hat{R}_n(g), \quad \text{with } \mathcal{G} \subset \mathcal{F}(\mathcal{X}, \mathcal{Y})$$

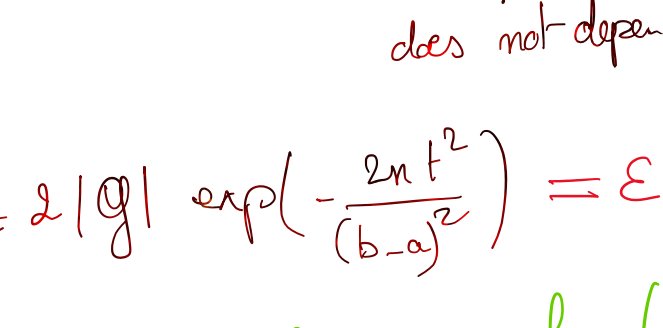
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Overfitting

Recall

$$\hat{g}_{n,G} \in \arg \min_{g \in \mathcal{G}} \hat{R}_n(g), \quad \text{with } \mathcal{G} \subset \mathcal{F}(\mathcal{X}, \mathcal{Y})$$

- Consider the regression problem with $\mathcal{G} = \mathcal{F}(\mathcal{X}, \mathcal{Y})$



- Several solution for $\hat{g}_{n,G}$
- Generalization issues : overfitting
- Need to restrict $\mathcal{G}$**

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Bias-variance decomposition

Define the oracle predictor over $\mathcal{G}$

$$g_{\mathcal{G},P}^* = \arg \min_{g \in \mathcal{G}} R_P(g)$$

- Then

$$R_P(\hat{g}_{n,G}) - R_P(g_{\mathcal{G},P}^*) = \underbrace{R_P(\hat{g}_{n,G}) - R_P(g_{\mathcal{G},P}^*)}_{\text{stochastic error}} = \underbrace{R_P(g_{\mathcal{G},P}^*) - R_P(g_{\mathcal{G},P}^*)}_{\text{estimation error}} - R_P(g_{\mathcal{G},P}^*)$$

- We focus on the stochastic error (estimation error relies on approximation theory)

- Fact 1** : trivial inequalities

$$R_P(\hat{g}_{n,G}) \geq R_P(g_{\mathcal{G},P}^*) \geq R_P(g_{\mathcal{G},P}^*), \quad \hat{R}_n(\hat{g}_{n,G}) \leq R_P(\hat{g}_{n,G})$$

- Fact 2** : Fundamental inequality

$$R_P(\hat{g}_{n,G}) - R_P(g_{\mathcal{G},P}^*) \leq 2 \sup_{g \in \mathcal{G}} |R_P(g) - R_P(\hat{g}_{n,G})|$$

- We called the excess-risk by empirical process.

$$R_P(\hat{g}_{n,G}) - R_P(g_{\mathcal{G},P}^*) = R_P(\hat{g}_{n,G}) - \hat{R}_n(\hat{g}_{n,G}) + \hat{R}_n(\hat{g}_{n,G}) - \hat{R}_n(g_{\mathcal{G},P}^*) + \hat{R}_n(g_{\mathcal{G},P}^*) - R_P(g_{\mathcal{G},P}^*) \leq 2 \sup_{g \in \mathcal{G}} |\hat{R}_n(g) - R_P(g)|$$

The finite case

Theorem

Assume $|\mathcal{G}| < +\infty$ and that there exist real numbers $a < b$ such that $a \leq \ell(y, y') \leq b$ for all $y, y' \in \mathcal{Y}$. Then $\forall \epsilon > 0$, with probability greater than $1 - \epsilon$

$$R_P(\hat{g}_{n,G}) - R_P(g_{\mathcal{G},P}^*) \leq (b - a) \sqrt{\frac{2 \ln(2|\mathcal{G}|) - 1}{n}}$$

Main argument for the proof :

Proposition (Hoeffding inequality)

Let $V_1, \dots, V_n$ be independent random variables such that $a_i \leq V_i \leq b_i$ be surely. Let $\bar{V}_n = \frac{1}{n} \sum_{i=1}^n V_i$. Then

$$P(|\bar{V}_n - \mathbb{E}[\bar{V}_n]| > t) \leq e^{-2nt^2 / \sum_{i=1}^n (b_i - a_i)^2} \quad \forall t > 0$$

- Question** : What if $|\mathcal{G}| = +\infty$?

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We have to control

$$P(R_P(\hat{g}_{n,G}) - R_P(g_{\mathcal{G},P}^*) > 2t)$$

$$\stackrel{\text{Fubini}}{\leq} P\left(\exists g \in \mathcal{G} : |\hat{R}_n(g) - R_P(g)| > t\right)$$

$$= P\left(\exists g \in \mathcal{G} : |\hat{R}_n(g) - R_P(g)| > t\right)$$

$$\stackrel{\text{union bound}}{\leq} \sum_{g \in \mathcal{G}} P(|\hat{R}_n(g) - R_P(g)| > t)$$

$$\leq \sum_{g \in \mathcal{G}} P\left(\frac{1}{n} \sum_{i=1}^n \ell(Y_i, g(X_i$$